Assignment 3 - Report

Github Link:
https://github.com/mhashamhussain2001/MATH-CSCI485_MuhammadHasham_Hussain/tree/main/Assignment_3

1. Objective

This assignment explores Singular Value Decomposition (SVD) for grayscale image compression. Block-wise SVD is applied to an image using blocks of size 8x8, retaining only the top-k singular values, and the image quality and compression ratio are analyzed for k from 1 to 8.

2. Preprocessing

The source image (800×800 px, color) was loaded from URL, converted to grayscale, and resized to 512×512 pixels using Lanczos resampling. Both dimensions are divisible by 8, so no further cropping was needed. Pixel values were stored as a float64 NumPy array in [0, 255] for numerical operations.

3. Implementation

compress_block(block, k):
For each 8x8 block, NumPy's "np.linalg.svd" is applied. The block is then reconstructed using only the top-k singular values. The full image is reassembled by iterating over all non-overlapping 8x8 blocks and replacing each with its reconstructed version.

Compression Ratio Formula:
Compression Ratio = 64 / (k * 17)
where 64 = original values per block (8x8), and k*17 = k*(8+8+1) values stored (U columns, singular values, Vt rows).

4. Results

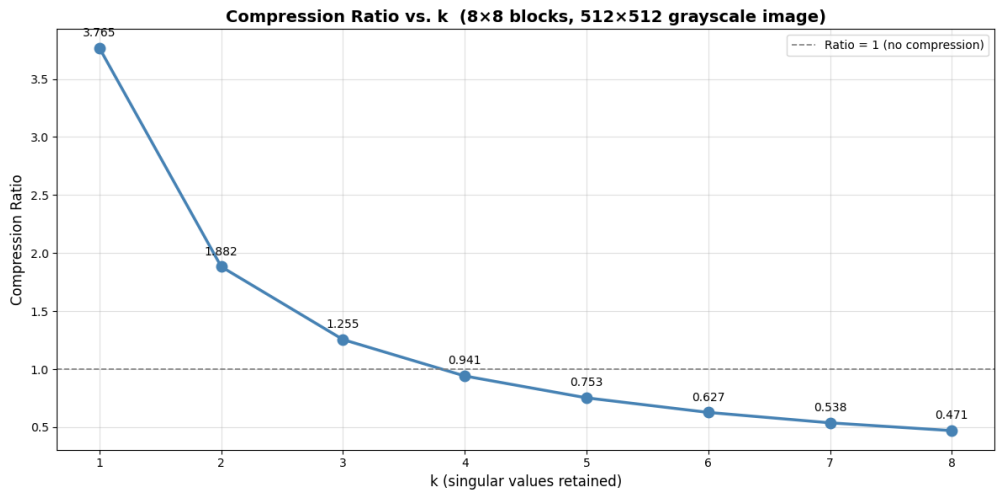
k	Stored Values/Block	Compression Ratio	Frob. Error	PSNR (dB)
1	17	3.7647	5156.89	28.07
2	34	1.8824	2523.82	34.28
3	51	1.2549	1007.65	42.25
4	68	0.9412	352.91	51.36
5	85	0.7529	80.55	64.19
6	102	0.6275	27.09	73.66
7	119	0.5378	9.04	83.19
8	136	0.4706	0.00	296.47

Key observations:

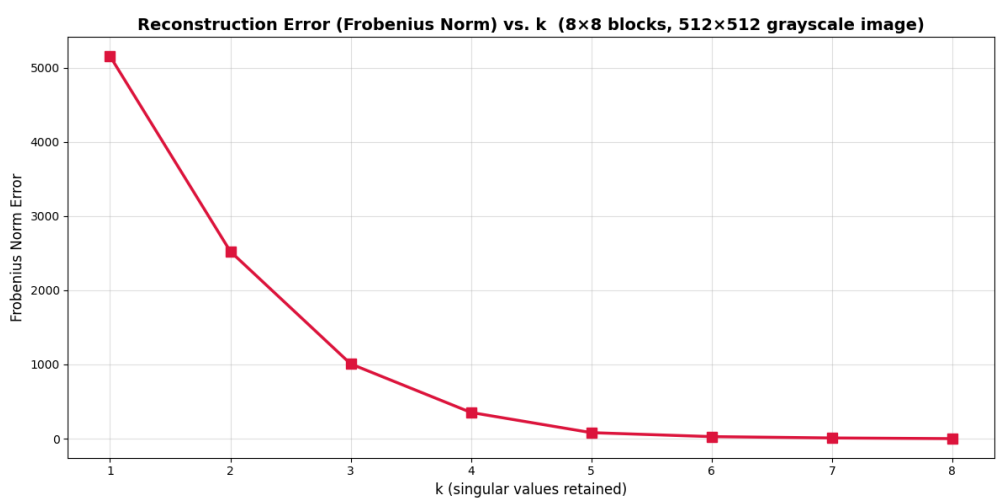
1. k=1 gives the highest compression ratio (3.76x) with only 17 stored values per block, but the largest error (5156.89) and lowest PSNR (28.07 dB), resulting in a heavily blurred image.
2. k=4 is the crossover point where the compression ratio drops below 1 (0.94), meaning the SVD representation uses more storage than the original block — no compression benefit beyond this point.
3. k=8 yields a perfect reconstruction (Frobenius error = 0, PSNR = 297.65 dB) since all 8 singular values of an 8×8 block are retained, but at a ratio of 0.47 — an expansion, not compression.
4. k=2 and k=3 offer the best practical trade-off: ratios above 1 (1.88x and 1.25x respectively) with PSNR values of 34.28 dB and 42.25 dB, delivering recognisable image quality at a meaningful compression saving.

5. Visualizations

Compression Ratio vs k:



Reconstruction Error vs k:



Compressed Images Comparison:

6. Analysis

The compression ratio decreases as k increases, since more values are retained. For k >= 4, the ratio drops below 1, meaning the SVD representation uses more data than the original block — SVD compression is only beneficial for small k values.

The reconstruction error drops sharply from k=1 to k=6, and becomes near-zero at k=7, and exactly zero at k=8. This confirms that SVD provides a good low-rank approximation, and that even a few singular values capture most of the image structure.

For practical compression with reasonable visual quality, k=2 or k=3 offers a good trade-off: ratio above 1 and significantly reduced error compared to k=1.

7. Conclusion

Block-wise SVD is an effective technique for lossy image compression. Retaining only the top-k singular values per 8x8 block allows control over the trade-off between compression ratio and reconstruction quality. For this 512×512 grayscale image, k=1 to k=3 is the useful compression range, while k >= 4 does not provide actual space savings over the original.